

THE COMPLETE PHYSICS ROADMAP

Penhe Observation se Nobel-Prize Research tak — Harvard · MIT · Oxford · Cambridge Level

Yeh PDF Physics seekhne ka **mukammal ordered roadmap** hai — bilkul pehli baar 'kya cheez girti kyun hai?' samajhne se lekar Quantum Field Theory, General Relativity, aur String Theory ke research frontier tak. Har topic pichle topic ki bunyaad par bana hai. Koi bhi cheez nahi chodi gayi — Newtonian Mechanics se Hawking Radiation tak sab yahan hai. Yeh roadmap Harvard, MIT, Oxford, Cambridge Physics ke syllabi par based hai.

LEVEL	STAGE	TARGET
Level 1	Pre-School (Age 3–6)	Observation, wonder, basic physical world
Level 2	Primary (Age 6–10)	Forces, energy, matter — concrete understanding
Level 3	Lower Secondary (Age 10–13)	Mechanics, waves, electricity — quantitative
Level 4	Upper Secondary (Age 13–16)	IGCSE Physics — all core topics
Level 5	Pre-University (Age 16–18)	A-Level / IB — calculus-based physics
Level 6	Undergraduate Yr 1–2 (18–20)	Classical physics — rigorous treatment
Level 7	Undergraduate Yr 2–3 (20–22)	Quantum mechanics, electrodynamics, stat mech
Level 8	Graduate / MSc (22–25)	Field theory, GR, condensed matter, particle physics
Level 9	PhD / Research (25+)	Frontier research — Nobel-level topics

1.1 Observation and Senses

- 1.1.1 Using 5 senses to explore the world
- 1.1.2 Describing objects: colour, size, shape, texture, weight
- 1.1.3 Sorting and classifying objects by properties
- 1.1.4 Comparing: heavier/lighter, taller/shorter, faster/slower
- 1.1.5 Noticing changes: ice melting, water drying
- 1.1.6 Asking 'why' and 'what if' questions

1.2 Forces and Motion — Intuitive

- 1.2.1 Pushing and pulling objects
- 1.2.2 Things that roll vs things that slide
- 1.2.3 Gravity: everything falls down
- 1.2.4 Ramps and slopes make things go faster
- 1.2.5 Magnets attract some materials
- 1.2.6 Floating and sinking

1.3 Light and Sound — Basic

- 1.3.1 Light comes from sources: sun, lamps, fire
- 1.3.2 Shadows are made when light is blocked
- 1.3.3 Mirrors reflect light
- 1.3.4 Sound is made by vibrations
- 1.3.5 Loud and soft, high and low sounds
- 1.3.6 Sound travels through air

1.4 Materials and Matter

- 1.4.1 Solids, liquids, gases — simple concept
- 1.4.2 Properties of materials: hard, soft, stretchy, waterproof
- 1.4.3 Materials change: cooking, freezing, melting
- 1.4.4 Natural vs man-made materials

1.5 Earth and Sky

- 1.5.1 Day and night cycle
- 1.5.2 The sun rises and sets
- 1.5.3 Weather: sunny, cloudy, rainy, windy
- 1.5.4 Seasons: hot and cold times of year
- 1.5.5 Stars visible at night
- 1.5.6 The Moon changes shape (phases — observation only)

★ *Parents: Experiments kar ke sikhao — ice cube se paani banana, magnet se paper clips uthana. Curiosity hi physics ka aaghaaz hai.*

2.1 Forces

- 2.1.1 Contact forces: push, pull, friction, air resistance
- 2.1.2 Non-contact forces: gravity, magnetism
- 2.1.3 Forces can change shape, speed, direction
- 2.1.4 Balanced and unbalanced forces (concept)
- 2.1.5 Weight vs mass (introductory distinction)
- 2.1.6 Friction: helpful and unhelpful
- 2.1.7 Air resistance and water resistance
- 2.1.8 Levers: fulcrum, effort, load
- 2.1.9 Pulleys and gears (simple concept)
- 2.1.10 Springs: stretching and compression

2.2 Motion

- 2.2.1 Speed = distance / time (concept with numbers)
- 2.2.2 Measuring distance and time
- 2.2.3 Faster and slower — comparing speeds
- 2.2.4 Direction of motion

2.3 Energy

- 2.3.1 Forms of energy: light, heat, sound, kinetic, potential, electrical, chemical
- 2.3.2 Energy transfers: candle wax → heat + light
- 2.3.3 Energy cannot be created or destroyed (concept)
- 2.3.4 Renewable and non-renewable energy sources
- 2.3.5 Saving energy — practical examples

2.4 Light

- 2.4.1 Light travels in straight lines
- 2.4.2 Light sources: luminous vs non-luminous
- 2.4.3 Transparent, translucent, opaque materials
- 2.4.4 Shadows: formation and size variation
- 2.4.5 Reflection: plane mirrors, angle of reflection
- 2.4.6 Refraction: straw in water appears bent
- 2.4.7 Colour: white light contains all colours (prism)
- 2.4.8 The spectrum: ROYGBIV

2.5 Sound

- 2.5.1 Sound produced by vibrations
- 2.5.2 Sound travels through solids, liquids, gases
- 2.5.3 Sound cannot travel through vacuum
- 2.5.4 Pitch: frequency of vibration
- 2.5.5 Loudness: amplitude of vibration
- 2.5.6 Echoes: reflection of sound
- 2.5.7 Speed of sound vs speed of light

2.6 Electricity

- 2.6.1 Simple circuits: battery, bulb, wire, switch
- 2.6.2 Conductors and insulators
- 2.6.3 Series and parallel circuits (conceptual)
- 2.6.4 Brightness of bulb vs number of batteries
- 2.6.5 Common electrical appliances and safety

2.7 Magnetism

- 2.7.1 Magnetic materials: iron, steel, nickel, cobalt
- 2.7.2 Poles: North and South
- 2.7.3 Attraction and repulsion
- 2.7.4 Magnetic field (concept — iron filings pattern)
- 2.7.5 Earth as a magnet — compass
- 2.7.6 Electromagnets

2.8 States of Matter

- 2.8.1 Solid, liquid, gas — properties and particle model
- 2.8.2 Melting, freezing, evaporation, condensation, sublimation
- 2.8.3 Heating and cooling cause changes of state
- 2.8.4 Melting and boiling points
- 2.8.5 Expansion and contraction with temperature

2.9 Earth and Space

- 2.9.1 Solar system: Sun and 8 planets
- 2.9.2 Earth orbits the Sun in one year
- 2.9.3 Moon orbits the Earth in one month
- 2.9.4 Phases of the Moon
- 2.9.5 Day and night: Earth's rotation
- 2.9.6 Seasons: Earth's tilt on its axis
- 2.9.7 Gravity holds planets in orbit
- 2.9.8 Stars, galaxies, the Milky Way — concept
- 2.9.9 Light year as a unit of distance

★ *Tip: NCERT Science Class 4-6 is level ke liye best hai. Simple experiments ghar mein karo.*

3.1 Measurements and Units

- 3.1.1 SI units: metre, kilogram, second, ampere, kelvin, mole, candela
- 3.1.2 Derived units: m/s, m/s², N, J, W, Pa, Hz, C, V, Ω
- 3.1.3 Scientific notation: 3.0×10^8 m/s
- 3.1.4 Significant figures and decimal places
- 3.1.5 Measuring instruments: ruler, vernier caliper, micrometer, stopwatch, thermometer, ammeter, voltmeter
- 3.1.6 Accuracy, precision, error, and uncertainty
- 3.1.7 Systematic and random errors
- 3.1.8 Reading scales and graphs

3.2 Kinematics — Motion in a Line

- 3.2.1 Distance vs displacement (scalar vs vector)
- 3.2.2 Speed vs velocity
- 3.2.3 Average speed = total distance / total time
- 3.2.4 Acceleration = change in velocity / time
- 3.2.5 Uniform acceleration: equations of motion
- 3.2.6 $s = ut + \frac{1}{2}at^2$
- 3.2.7 $v^2 = u^2 + 2as$
- 3.2.8 $v = u + at$
- 3.2.9 Distance-time graphs: gradient = speed
- 3.2.10 Velocity-time graphs: gradient = acceleration, area = distance
- 3.2.11 Free fall: $g = 9.8$ m/s² (or 10 m/s²)
- 3.2.12 Terminal velocity concept

3.3 Forces and Newton's Laws

- 3.3.1 Newton's First Law: inertia, objects at rest stay at rest
- 3.3.2 Newton's Second Law: $F = ma$
- 3.3.3 Newton's Third Law: equal and opposite reactions
- 3.3.4 Weight = mass $\times$ g
- 3.3.5 Free body diagrams
- 3.3.6 Resultant force
- 3.3.7 Friction force
- 3.3.8 Air resistance and drag
- 3.3.9 Tension, normal reaction, thrust
- 3.3.10 Pressure = Force / Area
- 3.3.11 Pressure in fluids: $P = h\rho g$
- 3.3.12 Archimedes' principle: buoyancy
- 3.3.13 Moments: moment = force $\times$ perpendicular distance
- 3.3.14 Principle of moments: sum of clockwise = sum of anticlockwise
- 3.3.15 Centre of gravity and stability

3.4 Work, Energy, Power

- 3.4.1 Work done = Force $\times$ distance (in direction of force)
- 3.4.2 Unit of work: Joule (J)
- 3.4.3 Kinetic energy: $KE = \frac{1}{2}mv^2$
- 3.4.4 Gravitational potential energy: $GPE = mgh$
- 3.4.5 Elastic potential energy
- 3.4.6 Conservation of energy
- 3.4.7 Energy transfers and transformation diagrams
- 3.4.8 Efficiency = (useful output / total input) $\times$ 100%
- 3.4.9 Power = Work done / time = $P = W/t$

- 3.4.10 Unit of power: Watt (W)
- 3.4.11 Power = Force $\times$ velocity

3.5 Thermal Physics

- 3.5.1 Temperature vs thermal energy
- 3.5.2 Thermometers and temperature scales: Celsius, Fahrenheit, Kelvin
- 3.5.3 Conversion: $K = ^\circ C + 273$
- 3.5.4 Internal energy — kinetic and potential energy of particles
- 3.5.5 Heat transfer: conduction, convection, radiation
- 3.5.6 Good and bad conductors
- 3.5.7 Convection currents in liquids and gases
- 3.5.8 Radiation: black bodies absorb and emit more
- 3.5.9 Specific heat capacity: $Q = mc\Delta T$
- 3.5.10 Latent heat: $Q = mL$ (change of state without temp change)
- 3.5.11 Specific latent heat of fusion and vaporisation
- 3.5.12 Thermal expansion of solids, liquids, gases
- 3.5.13 Anomalous expansion of water

3.6 Waves — Properties

- 3.6.1 Wave definition: energy transfer without matter transfer
- 3.6.2 Transverse vs longitudinal waves
- 3.6.3 Amplitude, wavelength, frequency, period
- 3.6.4 Wave equation: $v = f\lambda$
- 3.6.5 Reflection of waves
- 3.6.6 Refraction of waves
- 3.6.7 Diffraction of waves
- 3.6.8 Interference: constructive and destructive
- 3.6.9 Electromagnetic spectrum: radio, microwave, IR, visible, UV, X-ray, gamma
- 3.6.10 Properties and uses of each EM wave
- 3.6.11 Speed of light: 3×10^8 m/s

3.7 Sound Waves

- 3.7.1 Longitudinal wave — compressions and rarefactions
- 3.7.2 Speed of sound in different media
- 3.7.3 Frequency range: infrasound, audible, ultrasound
- 3.7.4 Ultrasound applications: sonar, medical imaging
- 3.7.5 Resonance — natural frequency
- 3.7.6 Stationary waves on strings
- 3.7.7 Doppler effect (qualitative)

3.8 Light — Geometric Optics

- 3.8.1 Law of reflection: angle of incidence = angle of reflection
- 3.8.2 Regular vs diffuse reflection
- 3.8.3 Image in plane mirror: virtual, upright, same size, laterally inverted
- 3.8.4 Refraction: Snell's law — $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- 3.8.5 Refractive index: $n = c/v$
- 3.8.6 Critical angle and total internal reflection
- 3.8.7 Optical fibres
- 3.8.8 Converging lens: focal point, focal length
- 3.8.9 Diverging lens
- 3.8.10 Ray diagrams for lenses
- 3.8.11 Lens formula: $1/f = 1/v + 1/u$
- 3.8.12 Magnification: $m = \text{image height} / \text{object height} = v/u$
- 3.8.13 Applications: eye, camera, telescope, microscope

3.8.14 Dispersion: prism splitting white light

3.9 Electricity and Circuits

- 3.9.1 Electric charge: positive and negative, coulombs
- 3.9.2 Electric current: $I = Q/t$, unit ampere
- 3.9.3 Conventional current direction
- 3.9.4 Potential difference (voltage): work done per unit charge, $V = W/Q$
- 3.9.5 Resistance: opposition to current, unit ohm
- 3.9.6 Ohm's law: $V = IR$
- 3.9.7 Ohmic and non-ohmic conductors
- 3.9.8 Series circuits: same current, voltages add, resistances add
- 3.9.9 Parallel circuits: voltages same, currents add, $1/R = 1/R_1 + 1/R_2$
- 3.9.10 Electrical power: $P = IV = I^2R = V^2/R$
- 3.9.11 Electrical energy: $E = Pt = IVt$
- 3.9.12 Cost of electricity: kWh
- 3.9.13 Fuses and circuit breakers — safety
- 3.9.14 Domestic wiring: live, neutral, earth

3.10 Magnetism and Electromagnetism

- 3.10.1 Magnetic field lines: direction and density
- 3.10.2 Field of a bar magnet
- 3.10.3 Magnetic effect of current: right-hand rule
- 3.10.4 Magnetic field of solenoid
- 3.10.5 Force on current in a magnetic field: $F = BIL$
- 3.10.6 Fleming's left-hand rule (motors)
- 3.10.7 DC electric motor: coil, commutator, brushes
- 3.10.8 Electromagnetic induction: Faraday's law
- 3.10.9 Fleming's right-hand rule (generators)
- 3.10.10 AC generator: slip rings
- 3.10.11 Transformer: $V_s/V_p = N_s/N_p$
- 3.10.12 Step-up and step-down transformers
- 3.10.13 Transmission of electricity: high voltage, low current

3.11 Radioactivity and Nuclear Physics (Introduction)

- 3.11.1 Atomic structure: protons, neutrons, electrons
- 3.11.2 Atomic number, mass number, isotopes
- 3.11.3 Radioactive decay: alpha, beta, gamma radiation
- 3.11.4 Properties of alpha, beta, gamma: charge, mass, penetration
- 3.11.5 Detection: Geiger-Muller tube, cloud chamber
- 3.11.6 Background radiation
- 3.11.7 Half-life: definition and calculations
- 3.11.8 Nuclear equations: balancing A and Z
- 3.11.9 Nuclear fission and chain reaction
- 3.11.10 Nuclear fusion
- 3.11.11 Applications: medical, nuclear power, dating
- 3.11.12 Hazards of ionising radiation

★ IGCSE Physics (Cambridge 0625): har chapter ke past papers solve karo. Graphs aur calculations dono practice karo.

4.1 Kinematics — 2D Motion

- 4.1.1 Vectors and scalars — full treatment
- 4.1.2 Adding vectors: triangle and parallelogram law
- 4.1.3 Resolving vectors into components
- 4.1.4 Projectile motion: horizontal and vertical components
- 4.1.5 Horizontal projectile: $x = ut$, $y = \frac{1}{2}gt^2$
- 4.1.6 Trajectory shape: parabola
- 4.1.7 Range, maximum height, time of flight
- 4.1.8 Projectile at an angle: launch velocity components
- 4.1.9 Relative velocity

4.2 Forces — Advanced

- 4.2.1 Newton's laws — quantitative problems
- 4.2.2 Friction: static and kinetic, coefficient of friction
- 4.2.3 Inclined plane problems
- 4.2.4 Atwood machine and connected bodies
- 4.2.5 Circular motion: centripetal acceleration $a = v^2/r$
- 4.2.6 Centripetal force: $F = mv^2/r$
- 4.2.7 Banked curves
- 4.2.8 Gravitational force: Newton's law of gravitation $F = Gm_1m_2/r^2$
- 4.2.9 Gravitational field strength $g = GM/r^2$
- 4.2.10 Orbital speed and period
- 4.2.11 Geostationary orbit
- 4.2.12 Kepler's Laws of planetary motion
- 4.2.13 Escape velocity: $v_{\text{esc}} = \sqrt{2GM/r}$
- 4.2.14 Hooke's Law: $F = kx$, spring constant
- 4.2.15 Simple Harmonic Motion — definition and basics

4.3 Energy and Momentum

- 4.3.1 Conservation of linear momentum: $p = mv$
- 4.3.2 Collisions: elastic and inelastic
- 4.3.3 Explosions and recoil
- 4.3.4 Impulse = force $\times$ time = change in momentum
- 4.3.5 Work-energy theorem
- 4.3.6 Conservation of mechanical energy
- 4.3.7 Power and efficiency — quantitative

4.4 Thermal Physics — Advanced

- 4.4.1 Kinetic theory of gases: assumptions
- 4.4.2 Pressure of a gas from particle collisions
- 4.4.3 Ideal gas laws: Boyle's, Charles's, Gay-Lussac's
- 4.4.4 Combined gas law: $PV/T = \text{constant}$
- 4.4.5 Ideal gas equation: $PV = nRT$
- 4.4.6 Avogadro's number, molar quantities
- 4.4.7 RMS speed of gas molecules
- 4.4.8 Absolute zero — particles have minimum KE
- 4.4.9 Internal energy and temperature relationship
- 4.4.10 First Law of Thermodynamics: $\Delta U = Q + W$
- 4.4.11 Thermodynamic processes: isothermal, adiabatic, isobaric, isochoric
- 4.4.12 Work done by gas: area under P-V graph
- 4.4.13 Second Law of Thermodynamics — concept

- 4.4.14 Heat engines and efficiency
- 4.4.15 Carnot cycle and maximum efficiency

4.5 Waves — Advanced

- 4.5.1 Principle of superposition
- 4.5.2 Standing (stationary) waves: nodes and antinodes
- 4.5.3 Standing waves on strings: harmonics and overtones
- 4.5.4 Standing waves in pipes: open and closed
- 4.5.5 Resonance in pipes and strings
- 4.5.6 Young's double-slit experiment: constructive and destructive interference
- 4.5.7 Fringe spacing: $w = \lambda D/d$
- 4.5.8 Single slit diffraction pattern
- 4.5.9 Diffraction grating: $d \sin \theta = n\lambda$
- 4.5.10 Polarisation of transverse waves
- 4.5.11 Malus's Law: $I = I_0 \cos^2 \theta$
- 4.5.12 Doppler effect (quantitative): $f' = f(v \pm v_{\text{observer}})/(v \mp v_{\text{source}})$
- 4.5.13 Applications of Doppler: astronomy, speed cameras, medical ultrasound

4.6 Electricity — Advanced

- 4.6.1 Electric field: $E = F/Q$
- 4.6.2 Field lines and equipotentials
- 4.6.3 Uniform electric field: $E = V/d$
- 4.6.4 Point charge field: $E = kQ/r^2 = Q/(4\pi\epsilon_0 r^2)$
- 4.6.5 Electric potential: $V = kQ/r$
- 4.6.6 Potential energy: $U = QV$
- 4.6.7 Motion of charges in electric fields
- 4.6.8 Capacitance: $C = Q/V$
- 4.6.9 Parallel plate capacitor: $C = \epsilon_0 A/d$
- 4.6.10 Energy stored in capacitor: $E = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/2C$
- 4.6.11 Capacitors in series and parallel
- 4.6.12 Charging and discharging: $Q = Q_0 e^{-(t/RC)}$, time constant $\tau = RC$
- 4.6.13 Electromotive force and internal resistance: $V = \mathcal{E} - Ir$
- 4.6.14 Resistivity: $R = \rho L/A$
- 4.6.15 Temperature dependence of resistance
- 4.6.16 Thermistor and LDR characteristics
- 4.6.17 Potential divider circuits
- 4.6.18 Wheatstone bridge
- 4.6.19 I-V characteristics: resistor, filament lamp, diode

4.7 Electromagnetism — Advanced

- 4.7.1 Magnetic flux: $\Phi = BA \cos \theta$
- 4.7.2 Faraday's law: $\mathcal{E} = -d\Phi/dt$
- 4.7.3 Lenz's law
- 4.7.4 Induced EMF in moving conductor: $\mathcal{E} = BLv$
- 4.7.5 Eddy currents and applications
- 4.7.6 Mutual inductance and self-inductance
- 4.7.7 Inductor: $L = N\Phi/I$
- 4.7.8 Energy stored in inductor: $E = \frac{1}{2}LI^2$
- 4.7.9 AC circuits: RMS values
- 4.7.10 Transformers — detailed treatment
- 4.7.11 Rectification: half-wave and full-wave
- 4.7.12 Smoothing with capacitors

4.8 Modern Physics — Quantum and Nuclear

- 4.8.1 Photoelectric effect: Einstein's explanation
- 4.8.2 Photon energy: $E = hf = hc/\lambda$
- 4.8.3 Work function and threshold frequency
- 4.8.4 Maximum kinetic energy: $E_k = hf - \Phi$
- 4.8.5 Wave-particle duality
- 4.8.6 de Broglie wavelength: $\lambda = h/mv$
- 4.8.7 Electron diffraction as evidence
- 4.8.8 Energy levels in atoms: Bohr model
- 4.8.9 Emission and absorption spectra
- 4.8.10 Energy level transitions: $E = hf$
- 4.8.11 X-ray production
- 4.8.12 Nuclear binding energy
- 4.8.13 Mass-energy equivalence: $E = mc^2$
- 4.8.14 Nuclear fission — energy release calculation
- 4.8.15 Nuclear fusion — stellar nucleosynthesis
- 4.8.16 Radioactive decay equations and series
- 4.8.17 Half-life calculations
- 4.8.18 Radioactive dating: carbon-14
- 4.8.19 Radiation dose and biological effects

★ *A-Level / IB Physics: Mathematics is essential. Calculus, vectors, and graphs — practice daily.*

5.1 Mechanics — Full Calculus Treatment

- 5.1.1 Kinematics using calculus: $v = dx/dt$, $a = dv/dt$
- 5.1.2 Integration to find displacement and velocity
- 5.1.3 Variable acceleration problems
- 5.1.4 Simple Harmonic Motion: $a = -\omega^2 x$
- 5.1.5 SHM equations: $x = A \cos(\omega t + \phi)$, $v = -A\omega \sin(\omega t + \phi)$
- 5.1.6 SHM energy: $E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2 A^2$
- 5.1.7 Damped oscillations: light, critical, heavy damping
- 5.1.8 Forced oscillations and resonance
- 5.1.9 Rotational motion: angular velocity ω , angular acceleration α
- 5.1.10 Torque: $\tau = I\alpha$
- 5.1.11 Moment of inertia: $I = \Sigma mr^2$
- 5.1.12 Rotational kinetic energy: $KE = \frac{1}{2}I\omega^2$
- 5.1.13 Angular momentum: $L = I\omega$, conservation
- 5.1.14 Gravitational field — full treatment with field lines and potential
- 5.1.15 Gravitational potential: $V = -GM/r$
- 5.1.16 Gravitational potential energy: $U = -GMm/r$

5.2 Thermal Physics — Statistical Approach

- 5.2.1 Internal energy: kinetic and potential energy of molecules
- 5.2.2 Boltzmann constant: $k_B = 1.38 \times 10^{-23} \text{ J/K}$
- 5.2.3 Mean KE of molecule: $\frac{1}{2}mv^2 = (3/2)k_BT$
- 5.2.4 Maxwell-Boltzmann speed distribution (qualitative)
- 5.2.5 Entropy — introduction: disorder and probability
- 5.2.6 Second Law in terms of entropy: $\Delta S \geq 0$
- 5.2.7 Carnot efficiency: $\eta = 1 - T_{\text{cold}}/T_{\text{hot}}$
- 5.2.8 P-V diagrams for thermodynamic cycles

5.3 Electric and Magnetic Fields — Full Treatment

- 5.3.1 Coulomb's law: $F = kQ_1Q_2/r^2$
- 5.3.2 Electric field strength: $E = F/Q = kQ/r^2$
- 5.3.3 Electric potential: $V = kQ/r$, work done $W = QV$
- 5.3.4 Capacitance — detailed derivation
- 5.3.5 Dielectrics and relative permittivity
- 5.3.6 Magnetic force on moving charge: $F = qv \times B$
- 5.3.7 Circular motion of charge in B field: $r = mv/qB$
- 5.3.8 Mass spectrometer principle
- 5.3.9 Hall effect and Hall voltage
- 5.3.10 Magnetic force on current: $F = BIL \sin\theta$
- 5.3.11 Biot-Savart law (qualitative)
- 5.3.12 Ampere's law (qualitative)
- 5.3.13 Flux linkage: $N\Phi = NBA \cos\theta$
- 5.3.14 Faraday's and Lenz's laws — full quantitative treatment
- 5.3.15 LC circuits and electromagnetic oscillations
- 5.3.16 Maxwell's displacement current (concept)
- 5.3.17 Electromagnetic waves as self-sustaining fields

5.4 Nuclear and Particle Physics

- 5.4.1 Nuclear model: protons, neutrons, electrons — sizes and densities
- 5.4.2 Strong nuclear force and its range
- 5.4.3 Binding energy per nucleon curve

- 5.4.4 Q-value of nuclear reactions
- 5.4.5 Radioactive decay law: $N = N_0 e^{(-\lambda t)}$
- 5.4.6 Activity: $A = \lambda N$
- 5.4.7 Half-life: $t_{1/2} = \ln 2 / \lambda$
- 5.4.8 Standard Model — particles: quarks, leptons, bosons
- 5.4.9 Quarks: up, down, strange, charm, bottom, top
- 5.4.10 Hadrons: baryons and mesons
- 5.4.11 Leptons: electron, muon, tau, neutrinos
- 5.4.12 Gauge bosons: photon, W, Z, gluon
- 5.4.13 Conservation laws: charge, baryon number, lepton number, strangeness
- 5.4.14 Feynman diagrams — introduction
- 5.4.15 Antiparticles and annihilation
- 5.4.16 Particle accelerators: cyclotron, synchrotron

5.5 Astrophysics

- 5.5.1 Stellar luminosity, flux, apparent and absolute magnitude
- 5.5.2 Stefan-Boltzmann law: $L = 4\pi R^2 \sigma T^4$
- 5.5.3 Wien's displacement law: $\lambda_{\text{max}} T = 2.90 \times 10^{-3} \text{ m}\cdot\text{K}$
- 5.5.4 Stellar spectra and classification: O B A F G K M
- 5.5.5 Hertzsprung-Russell diagram
- 5.5.6 Stellar evolution: main sequence, red giant, white dwarf, neutron star, black hole
- 5.5.7 Chandrasekhar limit
- 5.5.8 Hubble's Law: $v = Hd$
- 5.5.9 Cosmic redshift and expansion of universe
- 5.5.10 Big Bang theory and evidence: CMB radiation
- 5.5.11 Dark matter and dark energy (observational evidence)
- 5.5.12 Olbers' paradox
- 5.5.13 Cosmic microwave background

5.6 Quantum Physics — A-Level

- 5.6.1 Photoelectric effect — all experimental details
- 5.6.2 Photons and quantisation of energy
- 5.6.3 Wave-particle duality — full treatment
- 5.6.4 Electron diffraction and interference
- 5.6.5 Heisenberg's Uncertainty Principle: $\Delta x \Delta p \geq \hbar/2$
- 5.6.6 Quantum tunnelling — radioactive decay, STM
- 5.6.7 Laser principles: stimulated emission, population inversion
- 5.6.8 LED and photodiode

★ JEE Advanced, IPhO (International Physics Olympiad), Cambridge A-Level — highest quality preparation at this level.

6.1 Classical Mechanics I — Lagrangian Mechanics

- 6.1.1 Review of Newtonian mechanics with calculus
- 6.1.2 Generalised coordinates and degrees of freedom
- 6.1.3 Constraints: holonomic and non-holonomic
- 6.1.4 Virtual work and D'Alembert's principle
- 6.1.5 Lagrangian: $L = T - V$
- 6.1.6 Euler-Lagrange equations of motion
- 6.1.7 Cyclic coordinates and conserved momenta
- 6.1.8 Noether's theorem: symmetry $\leftrightarrow$ conservation law
- 6.1.9 Applications: simple pendulum, double pendulum, central force
- 6.1.10 Hamiltonian mechanics: $H = T + V$
- 6.1.11 Hamilton's equations of motion
- 6.1.12 Phase space
- 6.1.13 Poisson brackets
- 6.1.14 Canonical transformations
- 6.1.15 Hamilton-Jacobi equation
- 6.1.16 Action-angle variables
- 6.1.17 Liouville's theorem
- 6.1.18 Central force problem — full solution
- 6.1.19 Kepler problem: elliptical orbits from inverse square law
- 6.1.20 Two-body problem and reduced mass
- 6.1.21 Rigid body mechanics: Euler's equations
- 6.1.22 Inertia tensor
- 6.1.23 Euler angles
- 6.1.24 Precession, nutation, gyroscope
- 6.1.25 Coupled oscillators and normal modes

6.2 Electromagnetism I — Maxwell's Equations

- 6.2.1 Vector calculus: gradient, divergence, curl, Laplacian
- 6.2.2 Gauss's law in integral and differential form
- 6.2.3 Electrostatics: solving Laplace's and Poisson's equations
- 6.2.4 Method of images
- 6.2.5 Multipole expansion
- 6.2.6 Dielectrics: polarisation, displacement field D
- 6.2.7 Boundary conditions at interfaces
- 6.2.8 Magnetostatics: Biot-Savart law
- 6.2.9 Ampere's law in integral and differential form
- 6.2.10 Magnetic vector potential: $B = \nabla \times A$
- 6.2.11 Magnetic materials: magnetisation M , H field
- 6.2.12 Faraday's law in differential form: $\nabla \times E = -\partial B / \partial t$
- 6.2.13 Maxwell's displacement current: $\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \partial E / \partial t$
- 6.2.14 All four Maxwell's equations in differential form
- 6.2.15 Wave equation from Maxwell's equations
- 6.2.16 Electromagnetic waves: E and B fields, polarisation
- 6.2.17 Poynting vector and energy transport: $S = (1/\mu_0)(E \times B)$
- 6.2.18 Radiation pressure
- 6.2.19 Reflection and transmission at interfaces: Fresnel equations

6.3 Quantum Mechanics I

- 6.3.1 Failures of classical physics: blackbody, photoelectric, atomic spectra
- 6.3.2 de Broglie hypothesis

- 6.3.3 Wave function ψ and Born's probabilistic interpretation
- 6.3.4 Time-dependent Schrödinger equation: $i\hbar \partial\psi/\partial t = \hat{H}\psi$
- 6.3.5 Time-independent Schrödinger equation: $\hat{H}\psi = E\psi$
- 6.3.6 Infinite square well (particle in a box): exact solution
- 6.3.7 Finite square well
- 6.3.8 Quantum tunnelling: transmission coefficient
- 6.3.9 Quantum harmonic oscillator: ladder operators
- 6.3.10 Energy eigenvalues: $E_n = (n + \frac{1}{2})\hbar\omega$
- 6.3.11 Free particle: wave packets and group velocity
- 6.3.12 Heisenberg uncertainty principle: derivation
- 6.3.13 Operators, observables, eigenvalues
- 6.3.14 Expectation values: $\langle A \rangle = \int \psi^* \hat{A} \psi dx$
- 6.3.15 Commutators: $[x, p] = i\hbar$
- 6.3.16 Dirac notation (bra-ket)
- 6.3.17 Hilbert space and state vectors
- 6.3.18 Hydrogen atom: separation in spherical coordinates
- 6.3.19 Radial and angular equations
- 6.3.20 Quantum numbers: n, l, m_l
- 6.3.21 Spherical harmonics Y_l^m
- 6.3.22 Hydrogen energy levels: $E_n = -13.6/n^2 \text{ eV}$
- 6.3.23 Orbital shapes: s, p, d, f
- 6.3.24 Electron spin: $s = \frac{1}{2}, m_s = \pm\frac{1}{2}$
- 6.3.25 Spin operators and Pauli matrices
- 6.3.26 Addition of angular momentum: Clebsch-Gordan coefficients
- 6.3.27 Identical particles: bosons and fermions
- 6.3.28 Pauli exclusion principle
- 6.3.29 Perturbation theory: time-independent, first and second order
- 6.3.30 Degenerate perturbation theory
- 6.3.31 Time-dependent perturbation theory
- 6.3.32 Fermi's Golden Rule
- 6.3.33 Variational method
- 6.3.34 WKB approximation

6.4 Thermodynamics and Statistical Mechanics I

- 6.4.1 Thermodynamic state functions: U, H, S, G, F
- 6.4.2 Thermodynamic potentials and Maxwell relations
- 6.4.3 Chemical potential
- 6.4.4 Phase transitions and Clausius-Clapeyron equation
- 6.4.5 Van der Waals equation of state
- 6.4.6 Microstate and macrostate
- 6.4.7 Boltzmann entropy: $S = k_B \ln \Omega$
- 6.4.8 Canonical ensemble: partition function Z
- 6.4.9 Boltzmann distribution: $p_i \propto e^{-(E_i/k_B T)}$
- 6.4.10 Thermodynamic quantities from Z
- 6.4.11 Grand canonical ensemble and chemical potential
- 6.4.12 Classical ideal gas from statistical mechanics
- 6.4.13 Equipartition theorem
- 6.4.14 Quantum statistics: Fermi-Dirac distribution
- 6.4.15 Fermi energy and Fermi level
- 6.4.16 Bose-Einstein distribution
- 6.4.17 Bose-Einstein condensation
- 6.4.18 Blackbody radiation: Planck distribution derivation
- 6.4.19 Stefan-Boltzmann law from statistical mechanics

6.4.20 Debye model of heat capacity

6.4.21 Einstein model of heat capacity

6.5 Mathematical Methods for Physics

6.5.1 Linear algebra: matrices, eigenvalue problems

6.5.2 Differential equations: ODEs and PDEs in physics

6.5.3 Fourier series and transforms

6.5.4 Laplace transforms in physics

6.5.5 Complex analysis for physics: residue theorem, contour integrals

6.5.6 Green's functions

6.5.7 Sturm-Liouville theory

6.5.8 Special functions: Legendre polynomials, Bessel functions, Hermite, Laguerre

6.5.9 Tensor analysis: covariant and contravariant indices

6.5.10 Group theory in physics: symmetry groups

6.5.11 Variational calculus: Euler-Lagrange equations

6.5.12 Probability and statistics for experimental physics

★ MIT 8.03, 8.04, 8.044 — free OCW courses. Griffiths (EM + QM) aur Goldstein (Mechanics) ye level ke standard texts hain.

7.1 Quantum Mechanics II

- 7.1.1 Density matrix (density operator)
- 7.1.2 Entanglement and quantum non-locality
- 7.1.3 EPR paradox and Bell's inequalities
- 7.1.4 Bell's theorem and experimental tests
- 7.1.5 Quantum measurement problem and interpretations: Copenhagen, Many-Worlds, Pilot Wave
- 7.1.6 Path integral formulation: Feynman's approach
- 7.1.7 Scattering theory: cross-section, partial waves
- 7.1.8 Born approximation
- 7.1.9 Optical theorem
- 7.1.10 Relativistic quantum mechanics: Klein-Gordon equation
- 7.1.11 Dirac equation
- 7.1.12 Dirac matrices and spinors
- 7.1.13 Relativistic energy spectrum of hydrogen
- 7.1.14 Negative energy states and antimatter
- 7.1.15 Quantum information: qubits, quantum gates
- 7.1.16 Quantum entanglement and teleportation
- 7.1.17 Quantum computing — basics

7.2 Electrodynamics — Advanced

- 7.2.1 Radiation from accelerating charges
- 7.2.2 Larmor formula: $P = q^2 a^2 / 6\pi\epsilon_0 c^3$
- 7.2.3 Retarded potentials
- 7.2.4 Liénard-Wiechert potentials
- 7.2.5 Synchrotron radiation
- 7.2.6 Bremsstrahlung
- 7.2.7 Waveguides and cavity resonators
- 7.2.8 Coaxial transmission lines
- 7.2.9 Plasma physics introduction: Debye length, plasma frequency
- 7.2.10 Magnetohydrodynamics (MHD) — basics

7.3 Special Relativity — Full Treatment

- 7.3.1 Postulates of special relativity
- 7.3.2 Lorentz transformation
- 7.3.3 Time dilation: $\Delta t = \gamma \Delta t_0$
- 7.3.4 Length contraction: $L = L_0 / \gamma$
- 7.3.5 Relativistic velocity addition
- 7.3.6 Relativistic momentum: $p = \gamma m v$
- 7.3.7 Relativistic energy: $E = \gamma m c^2$
- 7.3.8 Rest energy: $E_0 = m c^2$
- 7.3.9 4-vectors and spacetime
- 7.3.10 Minkowski metric and spacetime interval
- 7.3.11 4-velocity, 4-momentum, 4-force
- 7.3.12 Relativistic dynamics
- 7.3.13 Covariant formulation of electrodynamics
- 7.3.14 Electromagnetic field tensor $F^{\mu\nu}$
- 7.3.15 Covariant Maxwell equations

7.4 General Relativity — Introduction

- 7.4.1 Equivalence principle: weak and strong

- 7.4.2 Gravitational time dilation
- 7.4.3 Curved spacetime concept
- 7.4.4 Riemannian geometry: metric tensor $g_{\mu\nu}$
- 7.4.5 Christoffel symbols and geodesics
- 7.4.6 Riemann curvature tensor
- 7.4.7 Ricci tensor and Ricci scalar
- 7.4.8 Einstein field equations: $G_{\mu\nu} = 8\pi G/c^2 T_{\mu\nu}$
- 7.4.9 Schwarzschild solution: metric outside spherical mass
- 7.4.10 Schwarzschild radius and black holes
- 7.4.11 Light deflection by gravity
- 7.4.12 Gravitational redshift
- 7.4.13 Perihelion precession of Mercury
- 7.4.14 Gravitational waves: production and detection
- 7.4.15 LIGO and GW150914 — first detection

7.5 Condensed Matter Physics I

- 7.5.1 Crystal structure: unit cell, Bravais lattices
- 7.5.2 Miller indices
- 7.5.3 X-ray diffraction: Bragg's law
- 7.5.4 Reciprocal lattice and Brillouin zones
- 7.5.5 Free electron model: Fermi energy, density of states
- 7.5.6 Band structure: energy bands, band gaps
- 7.5.7 Metals, insulators, semiconductors — band theory
- 7.5.8 Bloch's theorem
- 7.5.9 Tight-binding model
- 7.5.10 Nearly free electron model
- 7.5.11 Effective mass
- 7.5.12 Holes in semiconductors
- 7.5.13 Intrinsic and extrinsic semiconductors
- 7.5.14 p-n junction and diode
- 7.5.15 Transistor — brief
- 7.5.16 Phonons: quantum of lattice vibration
- 7.5.17 Specific heat from phonons
- 7.5.18 Superconductivity: Meissner effect, London equations, BCS theory — overview
- 7.5.19 High-temperature superconductors — phenomenology

7.6 Nuclear and Particle Physics

- 7.6.1 Nuclear models: liquid drop, shell model
- 7.6.2 Nuclear magic numbers
- 7.6.3 Beta decay: Fermi theory
- 7.6.4 Neutrino mass and oscillations
- 7.6.5 Weak interaction: parity violation
- 7.6.6 CP violation
- 7.6.7 Quark model of hadrons
- 7.6.8 Colour charge and confinement
- 7.6.9 QCD — overview of strong force
- 7.6.10 Electroweak unification: Glashow-Weinberg-Salam
- 7.6.11 Higgs mechanism and spontaneous symmetry breaking
- 7.6.12 Higgs boson — discovery 2012
- 7.6.13 Cosmic rays and astrophysical particles
- 7.6.14 Neutrino physics: solar, atmospheric, reactor
- 7.6.15 Beyond Standard Model — overview

7.7 Optics — Physical and Fourier

- 7.7.1 Coherence: temporal and spatial
- 7.7.2 Interferometry: Michelson, Mach-Zehnder, Fabry-Pérot
- 7.7.3 Laser physics: modes, cavities, gain
- 7.7.4 Gaussian beams
- 7.7.5 Fourier optics: spatial frequency
- 7.7.6 Diffraction theory: Fraunhofer and Fresnel
- 7.7.7 Imaging and resolution: Rayleigh criterion
- 7.7.8 Holography
- 7.7.9 Non-linear optics: second harmonic generation
- 7.7.10 Fibre optics and photonic crystals

★ *Landau & Lifshitz series, Jackson (EM), Shankar (QM), Misner-Thorne-Wheeler (GR) — ye graduate-level ke classic texts hain.*

8.1 Quantum Field Theory I

- 8.1.1 Why QFT? — relativistic quantum mechanics and multi-particle systems
- 8.1.2 Classical field theory: Lagrangian density
- 8.1.3 Canonical quantisation of scalar field (Klein-Gordon)
- 8.1.4 Creation and annihilation operators
- 8.1.5 Fock space
- 8.1.6 Normal ordering
- 8.1.7 Propagators and time-ordered products
- 8.1.8 Quantisation of Dirac field
- 8.1.9 Spin-statistics theorem
- 8.1.10 Quantisation of electromagnetic field: photons
- 8.1.11 Interaction picture and S-matrix
- 8.1.12 Wick's theorem
- 8.1.13 Feynman diagrams — derivation from QFT
- 8.1.14 Feynman rules for QED
- 8.1.15 Tree-level calculations in QED
- 8.1.16 Electron-muon scattering
- 8.1.17 Compton scattering
- 8.1.18 Pair production and annihilation
- 8.1.19 Cross sections and decay rates
- 8.1.20 Loop diagrams and divergences
- 8.1.21 Regularisation: dimensional, Pauli-Villars, cutoff
- 8.1.22 Renormalisation: mass, charge, field renormalisation
- 8.1.23 Running coupling constant
- 8.1.24 Renormalisation group equation
- 8.1.25 Callan-Symanzik equation
- 8.1.26 Beta function and asymptotic freedom

8.2 Quantum Field Theory II — Gauge Theories

- 8.2.1 Gauge invariance in electromagnetism
- 8.2.2 Non-Abelian gauge theories: Yang-Mills
- 8.2.3 QCD: SU(3) gauge theory of strong force
- 8.2.4 QCD Lagrangian
- 8.2.5 Asymptotic freedom in QCD
- 8.2.6 Confinement (phenomenological)
- 8.2.7 Perturbative QCD: parton model
- 8.2.8 Parton distribution functions
- 8.2.9 Deep inelastic scattering
- 8.2.10 Spontaneous symmetry breaking in field theory
- 8.2.11 Goldstone theorem
- 8.2.12 Higgs mechanism: mass generation for gauge bosons
- 8.2.13 Electroweak theory: $SU(2) \times U(1)$
- 8.2.14 W and Z boson masses
- 8.2.15 Weinberg angle
- 8.2.16 GIM mechanism and CKM matrix
- 8.2.17 Path integral formulation of QFT
- 8.2.18 Faddeev-Popov ghosts
- 8.2.19 BRST symmetry
- 8.2.20 Effective field theories
- 8.2.21 Operator product expansion

- 8.2.22 Anomalies: chiral and gauge anomalies
- 8.2.23 Instantons and non-perturbative effects
- 8.2.24 Solitons and topological defects

8.3 General Relativity — Advanced

- 8.3.1 Differential geometry: manifolds, tensors, differential forms
- 8.3.2 Covariant derivative and parallel transport
- 8.3.3 Geodesic equation derivation
- 8.3.4 Killing vectors and conserved quantities
- 8.3.5 Causal structure: null cones, past and future
- 8.3.6 Penrose diagrams
- 8.3.7 Schwarzschild black hole: coordinates and horizons
- 8.3.8 Kruskal extension and wormholes
- 8.3.9 Reissner-Nordström black hole (charged)
- 8.3.10 Kerr black hole (rotating)
- 8.3.11 Black hole thermodynamics: Bekenstein-Hawking entropy
- 8.3.12 Hawking radiation
- 8.3.13 Information paradox
- 8.3.14 Cosmological solutions: FRW metric
- 8.3.15 Friedmann equations
- 8.3.16 Cosmological constant and dark energy
- 8.3.17 Inflation: solving horizon and flatness problems
- 8.3.18 Primordial gravitational waves
- 8.3.19 Gravitational wave solution: linearised GR
- 8.3.20 GW production: binary systems, chirp mass

8.4 Condensed Matter Physics II

- 8.4.1 Second quantisation in condensed matter
- 8.4.2 Fermi liquid theory
- 8.4.3 BCS theory of superconductivity — full derivation
- 8.4.4 Ginzburg-Landau theory
- 8.4.5 Josephson effect
- 8.4.6 Integer quantum Hall effect
- 8.4.7 Fractional quantum Hall effect
- 8.4.8 Topological insulators and topological order
- 8.4.9 Berry phase and Berry curvature
- 8.4.10 Chern numbers and topological invariants
- 8.4.11 Weyl and Dirac semimetals
- 8.4.12 Graphene and Dirac physics in 2D
- 8.4.13 Strongly correlated systems: Mott insulator
- 8.4.14 Hubbard model
- 8.4.15 Kondo effect
- 8.4.16 Heavy fermions
- 8.4.17 Spin liquids
- 8.4.18 Renormalisation group in condensed matter
- 8.4.19 Critical phenomena and phase transitions
- 8.4.20 Scaling laws and universality
- 8.4.21 Conformal field theory in 2D condensed matter
- 8.4.22 Holography and condensed matter (AdS/CFT)

8.5 Statistical Mechanics — Advanced

- 8.5.1 Fluctuation-dissipation theorem
- 8.5.2 Linear response theory: Kubo formula

- 8.5.3 Non-equilibrium statistical mechanics
- 8.5.4 Stochastic processes: Langevin equation, Fokker-Planck
- 8.5.5 Ising model: exact solution in 1D, mean field in 2D
- 8.5.6 Monte Carlo methods
- 8.5.7 Molecular dynamics
- 8.5.8 Entropy production and arrow of time
- 8.5.9 Information theory and physics: Landauer's principle

8.6 Astrophysics and Cosmology — Graduate

- 8.6.1 Structure formation: power spectrum
- 8.6.2 CMB anisotropies: Sachs-Wolfe, acoustic peaks
- 8.6.3 Λ CDM model — parameters
- 8.6.4 Dark matter candidates: WIMPs, axions, sterile neutrinos
- 8.6.5 Dark energy: cosmological constant problem
- 8.6.6 Gravitational lensing: weak and strong
- 8.6.7 Neutron stars: equation of state, TOV equation
- 8.6.8 Pulsars and magnetars
- 8.6.9 Gamma-ray bursts
- 8.6.10 Active galactic nuclei and quasars
- 8.6.11 Galaxy formation and evolution
- 8.6.12 Nucleosynthesis: Big Bang and stellar
- 8.6.13 Multi-messenger astronomy: GW + EM

★ *Peskin & Schroeder (QFT), Wald (GR), Weinberg (QFT volumes), Ashcroft & Mermin (Condensed Matter) — PhD level standard texts.*

CHAPTER 9 — PhD / RESEARCH FRONTIER (Level 9 | Harvard · MIT · CERN · Caltech)

Yeh topics Nobel Prizes, Fields Medals (physics-related), aur world's top research labs — CERN, MIT, Caltech, Princeton IAS, Cambridge Cavendish, Oxford — ke active research areas hain. PhD candidates in areas mein original experimental ya theoretical results produce karte hain.

9.1 String Theory and M-Theory

- 9.1.1 Bosonic string theory: Nambu-Goto and Polyakov actions
- 9.1.2 Conformal invariance and Virasoro algebra
- 9.1.3 Critical dimension: 26 (bosonic), 10 (superstring)
- 9.1.4 Superstring theory: Type I, IIA, IIB, Heterotic
- 9.1.5 T-duality and S-duality
- 9.1.6 D-branes: Dirichlet boundary conditions
- 9.1.7 M-theory: 11-dimensional unification
- 9.1.8 Compactification and Calabi-Yau manifolds
- 9.1.9 Flux compactifications and the landscape
- 9.1.10 String landscape and the anthropic principle
- 9.1.11 Mirror symmetry
- 9.1.12 F-theory

9.2 AdS/CFT Correspondence

- 9.2.1 Anti-de Sitter space: geometry and symmetries
- 9.2.2 Conformal field theory: primary operators, OPE
- 9.2.3 Maldacena's conjecture: $\text{AdS}_5 \times \text{S}^5 \leftrightarrow \text{N}=4 \text{ SYM}$
- 9.2.4 GKPW prescription: bulk-boundary correspondence
- 9.2.5 Holographic renormalisation
- 9.2.6 Holographic entanglement entropy: Ryu-Takayanagi formula
- 9.2.7 Applications: holographic superconductors, strange metals
- 9.2.8 Black holes in AdS and thermodynamics
- 9.2.9 Bulk reconstruction and subregion duality

9.3 Quantum Gravity

- 9.3.1 Problems with quantising gravity: non-renormalisability
- 9.3.2 Loop quantum gravity (LQG): spin networks, spin foams
- 9.3.3 Causal dynamical triangulations
- 9.3.4 Asymptotic safety scenario
- 9.3.5 Black hole entropy and microstate counting
- 9.3.6 Information paradox: Page curve, island formula
- 9.3.7 ER = EPR conjecture (Maldacena-Susskind)
- 9.3.8 Holographic complexity
- 9.3.9 Quantum error correction in holography
- 9.3.10 Causal sets approach

9.4 Experimental High-Energy Physics

- 9.4.1 LHC experiments: ATLAS, CMS, LHCb, ALICE
- 9.4.2 Higgs boson measurements and couplings
- 9.4.3 Beyond Standard Model searches: SUSY, extra dimensions
- 9.4.4 Top quark physics
- 9.4.5 B-physics and CP violation measurements
- 9.4.6 Quark-gluon plasma: heavy-ion collisions
- 9.4.7 Future colliders: FCC, ILC, CLIC, muon collider
- 9.4.8 Neutrino experiments: NOvA, T2K, DUNE

- 9.4.9 Dark matter direct detection: XENON, LUX, PandaX
- 9.4.10 Dark matter indirect detection: gamma-ray telescopes
- 9.4.11 Axion searches: ADMX

9.5 Gravitational Wave Astronomy

- 9.5.1 LIGO/Virgo/KAGRA — detector physics
- 9.5.2 Binary black hole mergers — waveform modelling
- 9.5.3 Neutron star mergers: GW170817
- 9.5.4 Multi-messenger astronomy
- 9.5.5 Pulsar timing arrays: nanohertz GW background
- 9.5.6 LISA: space-based detector
- 9.5.7 Tests of GR with gravitational waves
- 9.5.8 Black hole spectroscopy

9.6 Quantum Computation and Quantum Information

- 9.6.1 Quantum circuits and universal gate sets
- 9.6.2 Quantum algorithms: Shor's, Grover's
- 9.6.3 Quantum error correction codes: stabiliser codes
- 9.6.4 Fault-tolerant quantum computing
- 9.6.5 Physical implementations: superconducting qubits, trapped ions, photonic
- 9.6.6 Quantum complexity theory: BQP, QMA
- 9.6.7 Quantum cryptography: BB84, device-independent QKD
- 9.6.8 Quantum networks and quantum internet
- 9.6.9 Quantum simulation: digital and analogue
- 9.6.10 Quantum thermodynamics
- 9.6.11 Quantum chaos and scrambling
- 9.6.12 Many-body localisation

9.7 Topological and Strongly Correlated Physics

- 9.7.1 Topological quantum field theories: Chern-Simons
- 9.7.2 Anyon statistics and non-Abelian anyons
- 9.7.3 Topological quantum computation
- 9.7.4 Topological semimetals: Weyl, Dirac
- 9.7.5 Floquet topological phases
- 9.7.6 Non-equilibrium topological phenomena
- 9.7.7 Quantum spin liquids: experimental candidates
- 9.7.8 High-T_c superconductivity: mechanism debates
- 9.7.9 Cuprate phase diagram
- 9.7.10 Strange metal behaviour and its theories
- 9.7.11 Moiré physics: twisted bilayer graphene superconductivity

9.8 Cosmology and Astrophysics Research Frontier

- 9.8.1 Inflationary models: single field, multifield, eternal inflation
- 9.8.2 Primordial gravitational waves: B-mode polarisation
- 9.8.3 Primordial non-Gaussianity
- 9.8.4 Baryon asymmetry and baryogenesis
- 9.8.5 Leptogenesis
- 9.8.6 Electroweak baryogenesis
- 9.8.7 Black hole shadow: Event Horizon Telescope
- 9.8.8 Fast radio bursts
- 9.8.9 Hubble tension — measurement and theoretical implications
- 9.8.10 21-cm cosmology
- 9.8.11 Stage-4 CMB experiments
- 9.8.12 Euclid, DESI, Rubin — large-scale structure surveys

9.9 Atomic, Molecular and Optical Physics

- 9.9.1 Laser cooling and trapping: Doppler cooling
- 9.9.2 Magneto-optical traps
- 9.9.3 Bose-Einstein condensates: creation and manipulation
- 9.9.4 Optical lattices and quantum simulation
- 9.9.5 Ultracold Fermi gases: BEC-BCS crossover
- 9.9.6 Precision spectroscopy: hydrogen 1S-2S transition
- 9.9.7 Optical atomic clocks
- 9.9.8 Tests of fundamental constants variation
- 9.9.9 Attosecond physics: electron dynamics
- 9.9.10 Strong field physics: above-threshold ionisation, HHG
- 9.9.11 Quantum optics: squeezed states, cavity QED
- 9.9.12 Circuit QED and superconducting quantum systems

9.10 Unsolved Problems in Physics

- 9.10.1 What is dark matter?
- 9.10.2 What is dark energy / cosmological constant?
- 9.10.3 Why is there more matter than antimatter?
- 9.10.4 How to reconcile QM and GR (quantum gravity)?
- 9.10.5 What causes high-temperature superconductivity?
- 9.10.6 Is there a theory of everything (ToE)?
- 9.10.7 What is the origin of ultra-high-energy cosmic rays?
- 9.10.8 What is the origin of mass of neutrinos?
- 9.10.9 Can the strong CP problem be solved (axion)?
- 9.10.10 Is the proton stable? (GUT prediction)
- 9.10.11 What happened before / at the Big Bang?
- 9.10.12 Is string theory / M-theory testable?
- 9.10.13 Arrow of time: why does entropy increase?
- 9.10.14 Measurement problem in quantum mechanics: which interpretation?
- 9.10.15 Do extra dimensions exist?

COMPLETE ROADMAP — AT A GLANCE

Chapter	Level	Age / Stage	Major Topics	Topics
1	L1	Age 3–6	Observation, forces, light, sound, matter	~35
2	L2	Age 6–10	Forces, energy, waves, electricity, space	~80
3	L3	Age 10–13	Mechanics, thermal, waves, EM, nuclear	~130
4	L4	Age 13–16	IGCSE: vectors, projectiles, capacitors, QM intro	~130
5	L5	Age 16–18	A-Level: SHM, rotational, fields, particle physics	~95
6	L6	Age 18–20	Lagrangian, Maxwell equations, QM I, stat mech	~145
7	L7	Age 20–22	QM II, GR intro, condensed matter, special rel	~105
8	L8	Age 22–25	QFT, gauge theories, advanced GR, topological CMP	~155
9	L9	Age 25+	String theory, AdS/CFT, GW astronomy, frontier	~130
TOTAL				~1005+ Topics

Is roadmap ko follow karo — Newton ke laws se Hawking radiation tak. Har topic pichle par bana hai. Physics mein kuch bhi magic nahi — sab logic aur mathematics hai. Mehnat karo, aur Harvard, MIT, Oxford ke darwaze khulenge.